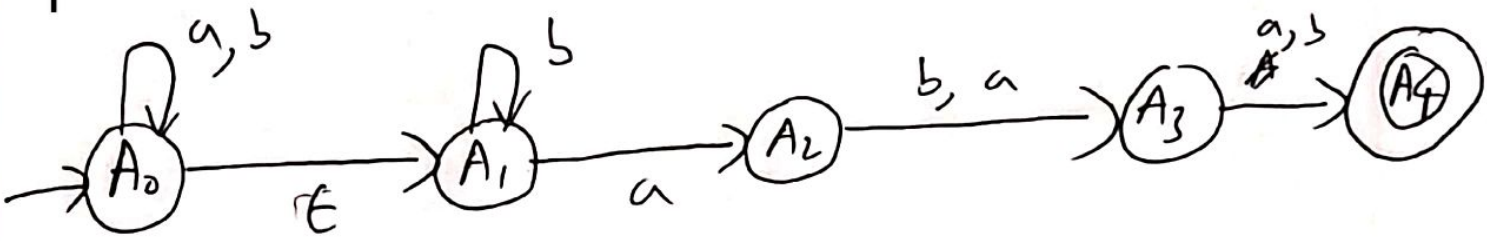
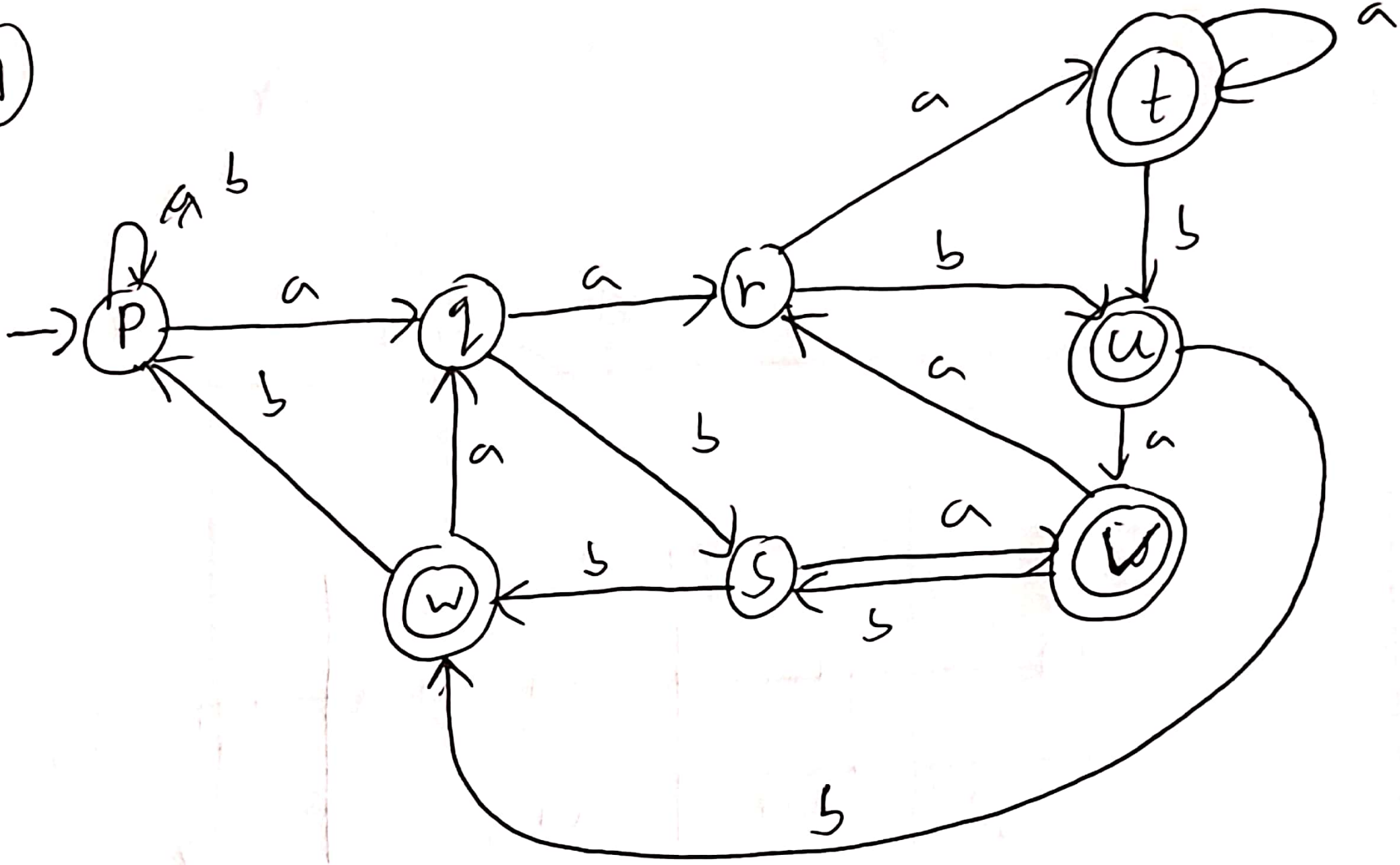
NFA/ε-NFA → DFA

1

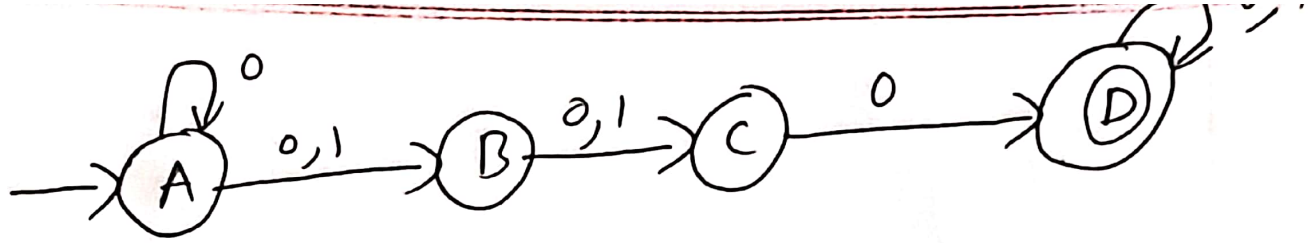


	a	b
$A_0, A_1 = P$	$A_0, A_1, A_2 = q$	$A_0, A_1 = P$
$A_0, A_1, A_2 = q$	$A_0, A_1, A_2, A_3 = r$	$A_0, A_1, A_3 = s$
$A_0, A_1, A_2, A_3 = r$	$A_0, A_1, A_2, A_3, A_4 = t$	$A_0, A_1, A_3, A_4 = u$
$A_0, A_1, A_3 = s$	$A_0, A_1, A_2, A_4 = v$	$A_0, A_1, A_4 = w$
$A_0, A_1, A_2, A_3, A_4 = t$	$A_0, A_1, A_2, A_3, A_4 = t$	$A_0, A_1, A_3, A_4 = u$
$A_0, A_1, A_3, A_4 = u$	$A_0, A_1, A_2, A_4 = (v)$	$A_0, A_1, A_4 = (w)$
$A_0, A_1, A_2, A_4 = v$	$A_0, A_1, A_2, A_3 = (r)$	A_0, A_1, A_3 (s)
$A_0, A_1, A_4 = w$	$A_0, A_1, A_2 = q$	$A_0, A_1 = P$

1

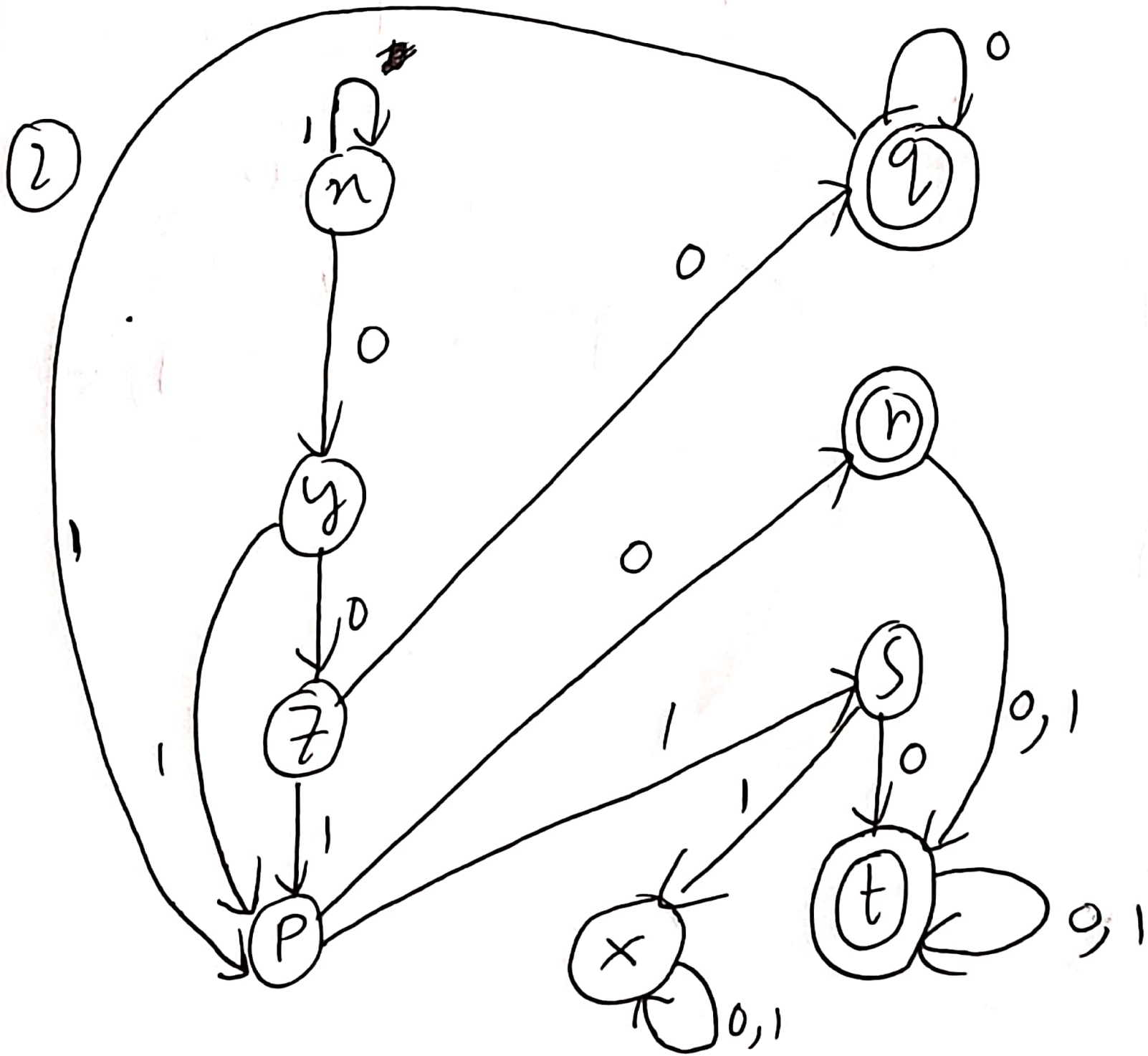


(2)

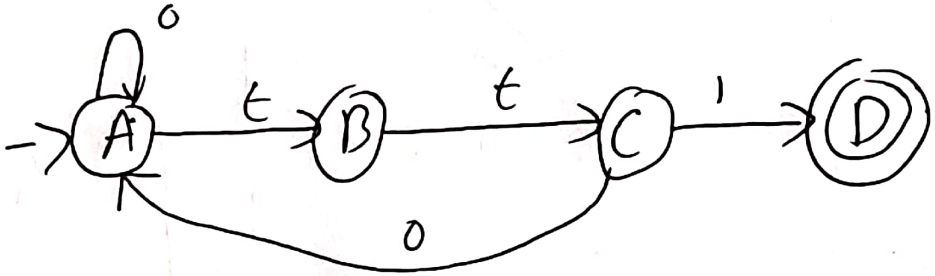


	0	1
A (x)	AB (y)	A (x)
A, B (y)	ABC (z)	BC (p)
A, BC (z)	ABCD (q)	BC (p)
BC (p)	CD (r)	C (s)
ABCD (q)	ABCD (q)	BC (p)
CD (r)	D (t)	D (t)
C (s)	D (t)	
D (t)	D (t)	D (t)

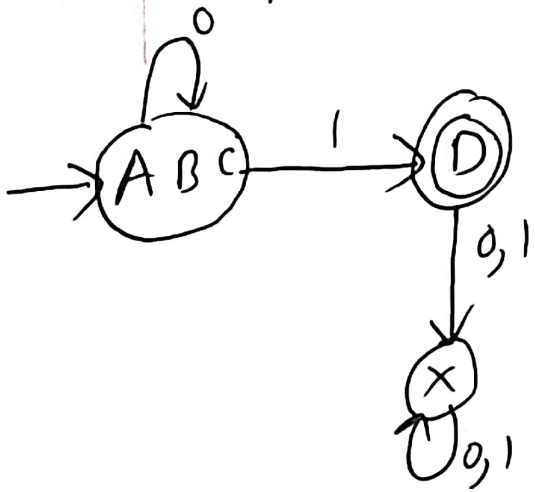
2



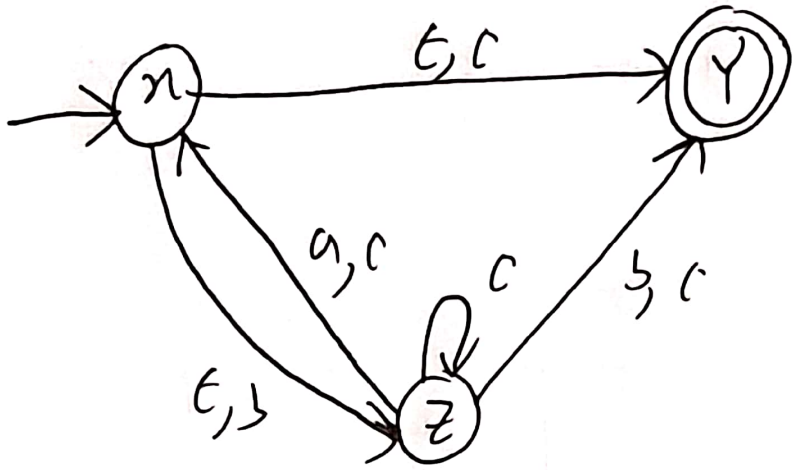
3



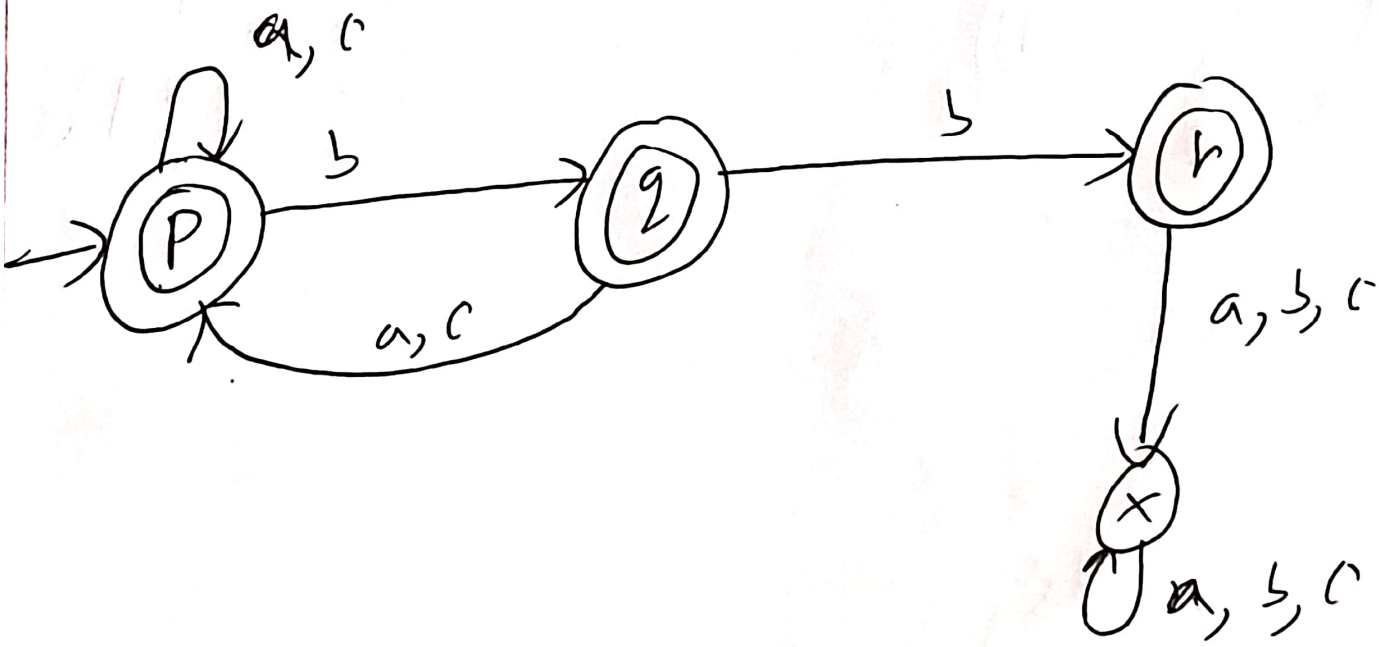
	0	1
ABC	ABC	D
D		



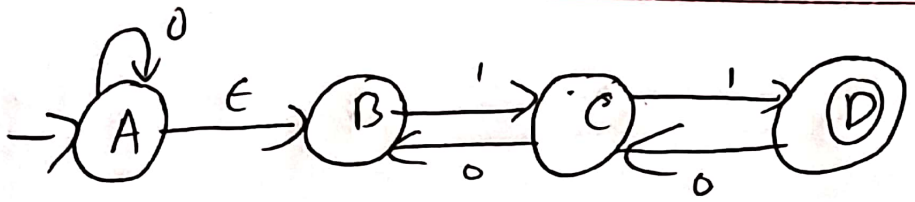
4)



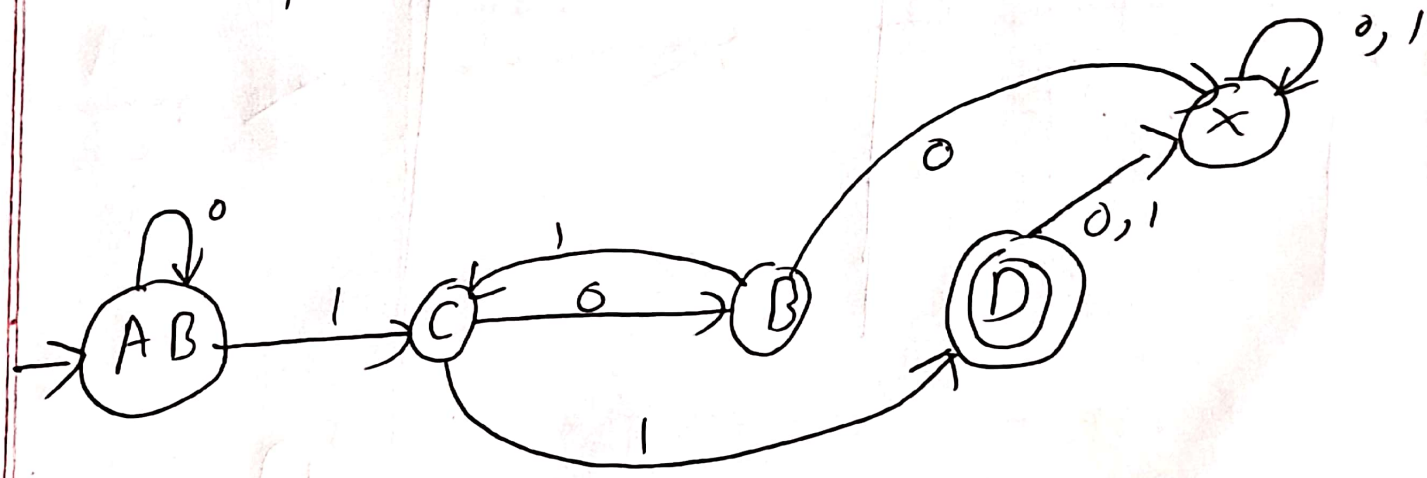
	a	b	c
xyz (P)	xyz (P)	Yz (q)	xyz (P)
Yz (q)	xYz (P)	Y (r)	xYz (P)
Y (r)			



(5)



	0	1
AB	AB	C
C	B	D
B	..	C
D		



(b) $\pi_1 = (A, B, D, E, F, G, H)\ (C)$

G_1 $\quad$ G_2

For a transition

$A \to G_1$
$B \to G_1$
$C \to G_1$
$D \to G_2$
$E \to G_1$
$F \to G_2$
$G \to G_1$
$H \to G_1$

For b transition

$A \to G_1$
$B \to G_2$
$C \to G_2$
$D \to G_1$
$E \to G_1$
$F \to G_1$
$G \to G_1$
$H \to G_2$

$\pi_2 = (A, E, G)\ (B, H)\ (D, F)\ (C)$

G_1 $\quad$ G_2 $\quad$ G_3 $\quad$ G_4

For a transition

$A \to G_2$
$B \to G_1$
$C \to G_1$
$D \to G_4$
$E \to G_2$
$F \to G_4$
$G \to G_2$
$H \to G_1$

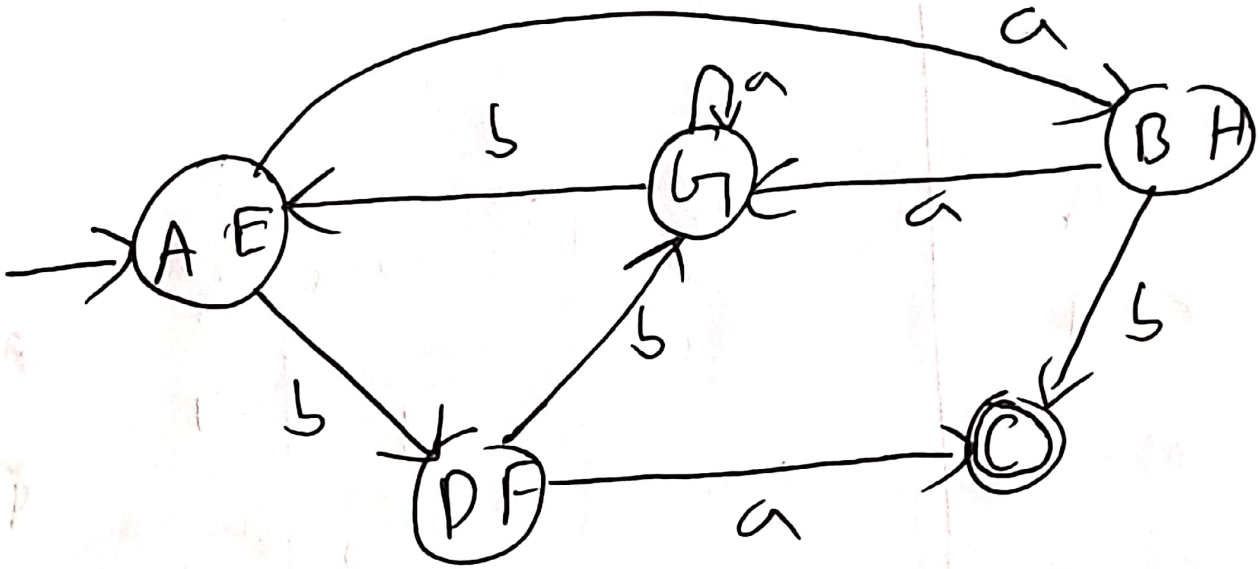
For b transition

$A \to G_3$
$B \to G_4$
$C \to G_4$
$D \to G_2$
$E \to G_3$
$F \to G_1$
$G \to G_1$
$H \to G_4$

π_3 $\quad (A, E)\ (G)\ (B, H)\ (D, F)\ (C)$

~~tro~~

no more partition possible.



7

$\pi_1 = (A, B, C, E, F)$ (D)

G_1 G_2

for 0	for 1
$A \to G_1$	$A \to G_2$
$B \to G_1$	$B \to G_2$
$C \to G_1$	$C \to G_1$
$D \to G_1$	$D \to G_1$
$E \to G_2$	$E \to G_2$
$F \to G_1$	$F \to G_1$

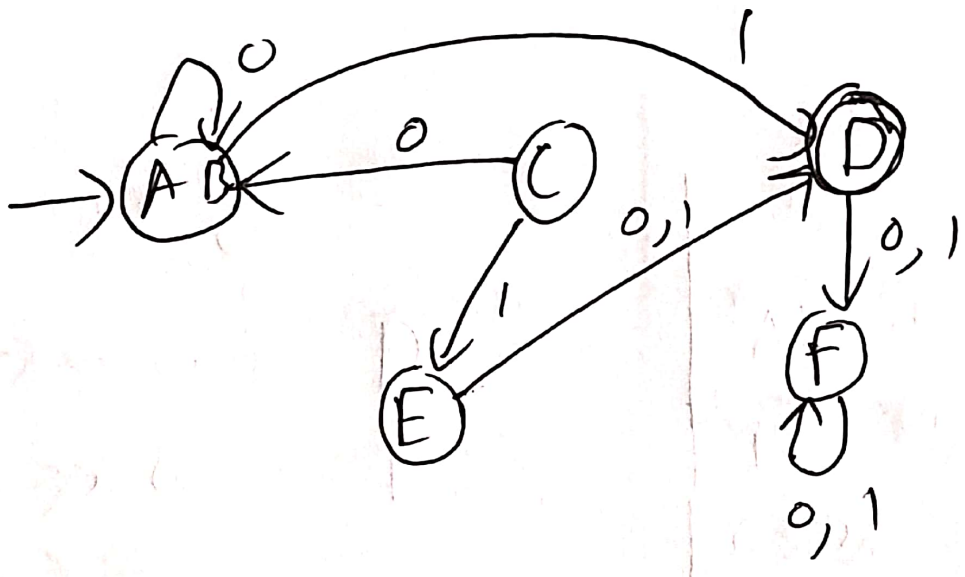
$\pi_2 = (A, B)$ (C, F) (E) (D)

G_1 G_2 G_3 G_4

For 0	For 1
$C \to G_1$	$C \to G_3$
$F \to G_2$	$F \to G_2$

rest (A, B) (E) (D) not shown as they remain same.

$\pi_3 = (A, B) \quad (C) \quad (D) \quad . (E) \quad (F)$



8.

$\pi_1 = (A, B, C, D)$ (E)

G_1 G_2

For a

$A \to G_1$ $D \to G_1$

$B \to G_1$ $E \to G_1$

$C \to G_1$

For b

$A \to G_1$ $D \to G_2$

$B \to G_1$ $E \to G_1$

$C \to G_1$

$\pi_2 = (A, B, C)$ (D) (E)

G_1 G_2 G_3

For a

$A \to G_1$ $D \to$

$B \to G_1$ $E \to$

$C \to G_2$

For b

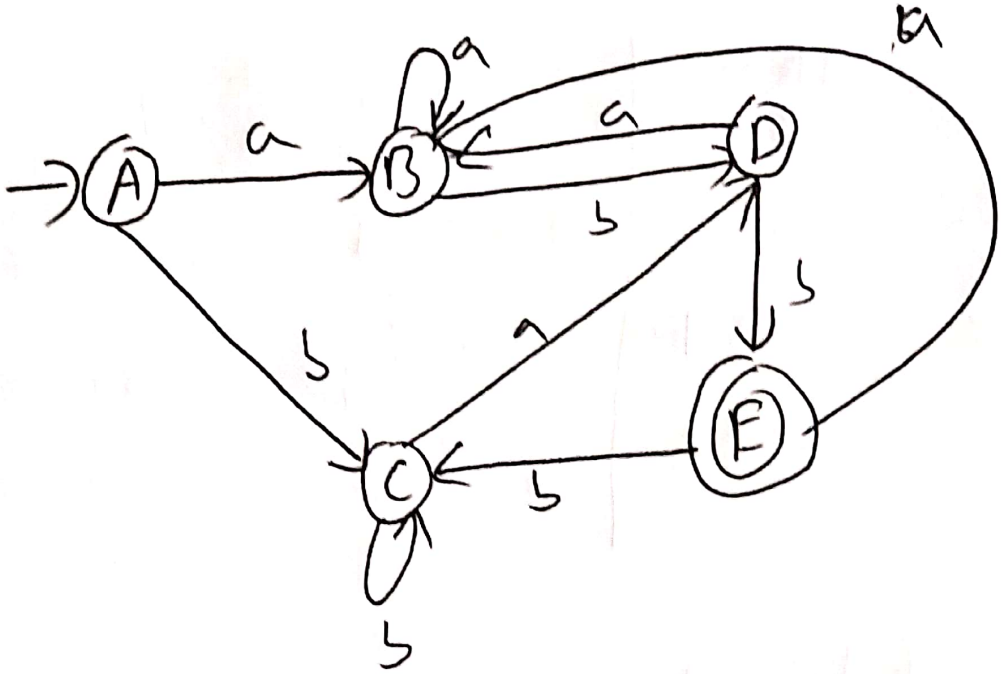
$A \to G_1$ $D \to$

$B \to G_2$ $E \to$

$C \to G_1$

D, E not shown as ~~remain same~~ already different.

π_3 : (A) (B) (C) (D) (E)



9. $\pi_1 = (A, B, C) \quad (D, E)$

$G_1 \qquad G_2$

For, 0		For 1	
$A \to G_1$	$D \to G_2$	$A \to G_1$	$D \to G_2$
$B \to G_1$	$E \to G_2$	$B \to G_2$	$E \to G_1$
$C \to G_1$		$C \to G_1$	

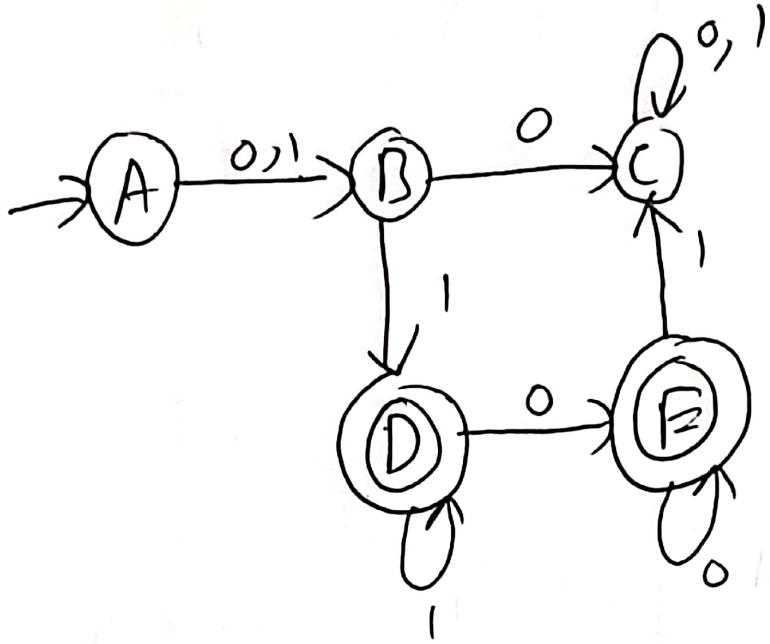
$\pi_2 = \cancel{(A)\ E} \quad (A, C) \quad (B) \quad (D) \quad (E)$

$G_1 \quad G_2 \quad G_3 \quad G_4$

For 0	For, 1
$A \to G_2$	$A \to G_2$
$C \to G_1$	$C \to G_1$

no more partitions

$\pi_3 = (A)(B)\ (C)\ (D)\ (E)$


A
B
C
D
E
0,1
0
0,1
1
1
0
1
0

10

$\pi_1 = (A, B, C, E, F)$ (D, G)

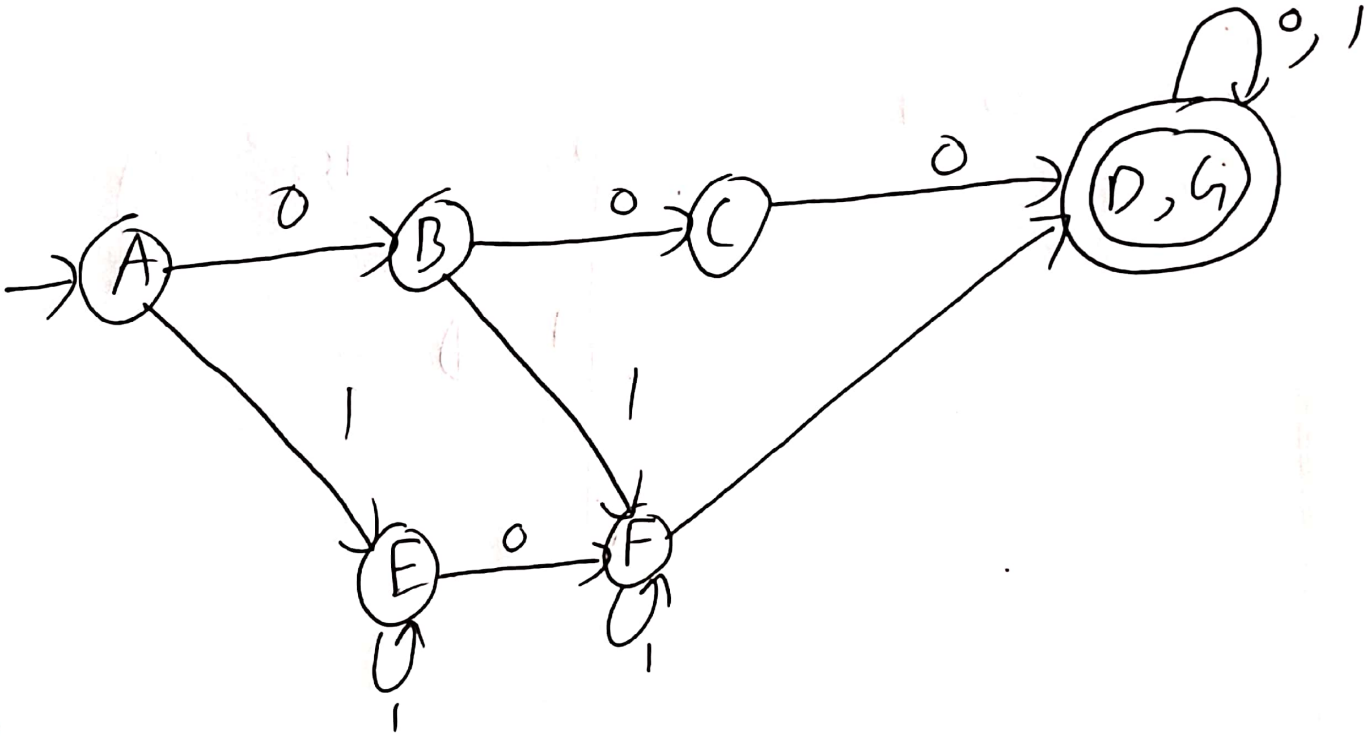
G_1 G_2

For 0	For 1
$A \to G_1$	$A \to G_1$
$B \to G_1$	$B \to G_1$
$C \to G_2$	C
$D \to G_2$	$D \to G_2$
$E \to G_1$	$E \to G_1$
$F \to G_2$	$F \to G_1$
$G \to G_2$	$G \to G_2$

$\pi_2 = (A, B, E)$ (C) (D, G) (F)

G_1 G_2 G_3 G_4

For 0,	For, 1
$A \to G_1$	$A \to G_1$
$B \to G_2$	$B \to G_4$
$E \to G_4$	$E \to G_1$

rest not shown as they don't change.

$\pi_3 = (A)\ (B)\ (C)\ \cancel{(D)}\ (D,G)\ (E)\ (F)$